

Time and Resource Log

Total Time Spent

Data Exploration: 6 Hours

Literature Reading: 5 Hours

Implementation: 14 Hours

Training: 8 Hours

Evaluation: 4 Hours

Report Writing: 5 Hours

Machine Used

Device: MacBook Air (Apple Silicon)

Framework: PyTorch with MPS backend

GPU: Apple Integrated GPU

VRAM: Shared memory architecture

Training Details

Epochs: 15

Approximate training time per epoch: 8–12 Minutes

Total wall-clock training time: ~3 Hours

Constraints

Severe dataset imbalance causing dominant background predictions

SAR image speckle noise affecting segmentation quality

Local hardware limitations compared to dedicated GPU servers

TIFF compatibility warnings during preprocessing and visualization

Limited experimentation time preventing extensive hyperparameter tuning and long training cycles

Sparse damaged regions making positive sample learning difficult

Multimodal EO-SAR feature differences increasing fusion complexity